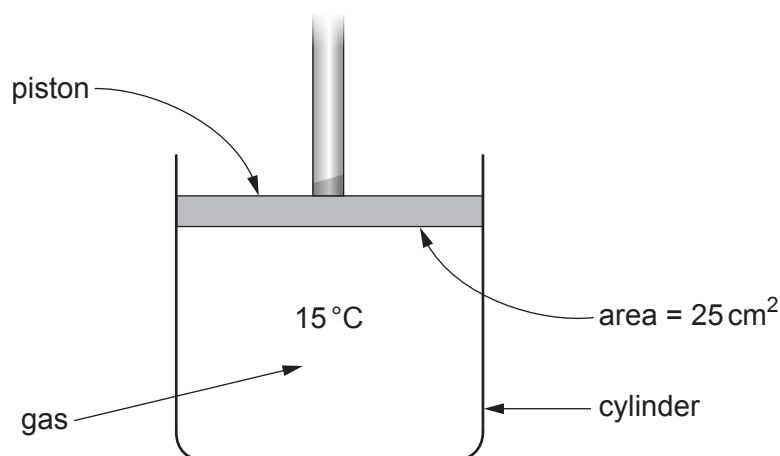


10. A piston holds 125 cm^3 of gas in a cylinder at a temperature of 15°C .
The surface area of the lower surface of the piston is 25 cm^2 .
The gas exerts a force of 250 N on this surface.



Use the information above to answer the following questions.

- (a) Use an equation from page 2 to calculate the pressure of the gas inside the cylinder. [2]

pressure = N/cm^2

- (b) The temperature of the gas increases to 298 K and the gas is compressed so the volume becomes 75 cm^3 .

- (i) Convert **15°C** into K .
Use the equation:

[1]

$$T/\text{K} = \theta/^\circ\text{C} + 273$$

$T = \text{..... K}$



(ii) Use the equation:

$$\frac{pV}{T} = \text{constant}$$

to calculate the new gas pressure.

[4]

pressure = N/cm²

Examiner
only

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END OF PAPER

